

IRexp experimental infrared band lists from the PMC Open Access Subset and Chemotion

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Abstract

IRexp is a redistributable collection of **experimental infrared band lists** (cm⁻¹ peak positions) mined from open chemistry literature, optionally with ¹H/¹³C shifts and resolved structures. The release holds **121,233** records (119,345 PMC OA; 1,888 Chemotion/RADAR4Chem), with **43,060** structure-linked and **33,201** full IR + ¹H + ¹³C + structure quadruples. IRexp stores **numeric band lists**, not absorbance traces. Records carry `source_doi` and stamped `license / license_pool`. Licensing is **mixed**: Europe PMC join yields **87,617** commercial (CC-BY/CC0), **20,938** non-commercial (NC*), **10,781** empty/unknown (excluded from commercial Zenodo), and **1,897** ShareAlike (Chemotion + rare PMC SA) — see `docs/scientific_data/LICENCE_REMEDIATION.md`. Reuse: multimodal training and complementary elucidation benchmarks citing this Descriptor. Data: Hugging Face `ilkhamfy/IRexp`; Zenodo [TODO: 10.5281/zenodo.XXXXXXX]. Code is MIT-licensed.

Background & Summary

Infrared (IR) spectroscopy is routine in organic characterisation, yet **open, redistributable** collections of *experimental* IR data remain sparse relative to modern machine-learning needs. Digitised absorbance libraries such as the NIST Chemistry WebBook¹ and AIST SDBS² are valuable but either modest in size or **view-only** (no bulk redistribution). Computational IR–NMR resources published in *Scientific Data* expand multimodal coverage with simulated spectra³, while large literature mines for **NMR** peak lists (notably NMRexp⁴) demonstrate that peer-reviewed experimental spectral *lists* with DOI traceability are in scope for this journal. Concurrent literature corpora that include IR among other modalities (for example NMRSpec/NMRTrans⁵) further motivate an **IR-focused, redistributable** band-list resource rather than another closed spectrum archive.

IRexp fills a specific wedge. Experimental sections of chemistry papers conventionally report per-compound **IR band lists** (wavenumbers in cm⁻¹) together with ¹H/¹³C NMR shift lists. That textual convention is the object language models and many elucidation pipelines consume, and it is a **different object** from a digitised spectrum. IRexp therefore:

1. Harvests open-access full text from the **PMC Open Access Subset** bulk S3 mirror and peak lists from the **Chemotion** FT-IR deposit.
2. Extracts deterministic numeric IR band lists (and co-reported NMR strings where present).
3. Resolves compound names to canonical structures with OPSIN⁶, RDKit⁷, and SELFIES⁸ where possible.

4. Releases **extracted numbers only** — no PDFs, figures, or article full text — with source accessions for attribution.

Among openly redistributable *text-derived IR band lists*, IRexp is large by record count (121,233). It does not claim to replace SDBS or NIST absorbance libraries; SDBS alone holds more structure-linked *spectra* than IRexp holds structure-linked band lists. The intended reuse is multimodal pre-training, supervised IR→structure modelling on `irexp_resolved`, and as the literature substrate for complementary elucidation benchmarks described elsewhere (forthcoming ICLR manuscript on IRSpectra-Bench; cite that work for protocol and model results — **not** reproduced here).

What this Data Descriptor does not contain. No hypothesis tests, no large-language-model accuracy tables, and no stage-decomposition of elucidation performance. Those belong in the complementary research paper.

Methods

Source corpora

PMC Open Access Subset. Primary harvest uses the NCBI PMC OA bulk distribution on Amazon S3 (`s3://pmc-oa-opendata`, HTTPS endpoint `https://pmc-oa-opendata.s3.amazonaws.com`)⁹. A harvest snapshot records **188,016** distinct PMC identifiers scanned (`data/irexp/seen_papers.txt.gz`). The released corpus retains records from **15,416** unique `PMC:*` accessions. Extraction operates on open-access plain text / JATS available through that OA path (and Europe PMC full-text XML for validation re-fetches). The **released** IRexp DOIs are exclusively `PMC:*` accessions or the Chemotion deposit DOI; no paywalled publisher DOI appears in the 121,233-record file.

PMC OA is **not** a single licence. The subset is partitioned into commercial-use, non-commercial, and other packages⁹. Every unique PMCID in IRexp (15,416) was joined to Europe PMC `license` metadata (`scripts/join_pmc_licences.py`); each record carries `license` / `license_pool`. Empty/unknown licences are **excluded** from the commercial Zenodo pool (`data/NOTICE`; `docs/scientific_data/LICENCE_REMEDIATION.md`).

Chemotion / RADAR4Chem. **1,888** records come from the Chemotion Repository FT-IR collection deposited at RADAR4Chem (DOI 10.22000/0GoEQGlsZGElrgst)¹⁰, licensed **CC-BY-SA-4.0**. These rows are **peak lists derived from deposited experimental FT-IR spectra** (not author-transcribed experimental-section prose). They carry a stamped `license=CC-BY-SA-4.0` and `source_doi` equal to the deposit DOI. Median band count is **39** versus **9** for PMC-transcribed lists; users modelling band density should treat the pools separately.

Excluded. AIST SDBS (view-only; no bulk export)²; NIST WebBook join code exists in the repository but contributes **0** records to the released IRexp (`ir_source` is always `experimental` for released rows).

Discovery and fetch (released corpus)

1. Enumerate PMC OA identifiers / packages from the S3 OA mirror.
2. Fetch OA full text for experimental-section mining.
3. Ingest Chemotion deposit peak lists with structure metadata from the RADAR4Chem release.
4. Persist harvest provenance in `seen_papers.txt.gz` and optional rawer rows in `ir_harvest_snapshot.jsonl` (134,893 rows including intermediate prose fields used to diagnose materials-text false positives; the curated release is `irexp.jsonl.gz`).

Development adapters for additional publishers exist in the codebase for exploration; they are **outside the construction path of the released dataset**. Methods for IReXP as published here are restricted to **PMC-OA S3 + Chemotion**.

Extraction (band lists, not spectra)

A deterministic regular-expression pipeline (`spectro_scraper/extract.py`) segments experimental text per compound and extracts:

- IR wavenumbers → `ir_bands_cm-1` (list of floats, cm^{-1});
- ^1H and ^{13}C payloads → `h_nmr` / `c_nmr` (author strings, when present).

Gates reject scan-range artefacts and common prose false positives (for example hydrogel / materials narrative mistaken for IR lists). **No curve digitisation** is performed: figures are not traced; JCAMP-DX is not stored.

Structure resolution

Where an IUPAC or systematic name is available, names are converted with OPSIN (py2opsin)⁶, canonicalised with RDKit⁷, and encoded as InChIKey and SELFIES⁸. An optional PubChem¹¹ name fallback (USE_PUBCHEM) handles trivial names. Structure coverage of the full release is **35.5%** (43,060 / 121,233). The structure-complete split is shipped as `irexp_resolved`.

Licence handling

Pool	Records	Stamp
commercial (CC-BY + CC0)	87,617	<code>license_pool=commercial</code> — Zenodo / Sci Data primary
non_commercial (CC-BY-NC*)	20,938	held aside
sharealike (Chemotion + rare PMC SA)	1,897	CC-BY-SA / CC-BY-SA-4.0
empty_unknown	10,781	excluded from commercial deposit

`scripts/join_pmc_licences.py` stamps every row; `scripts/split_license_pools.py` reports provenance (`pool_of`) and materialises pool files under `data/irexp/licence_pools/`. Narrative and policy: `docs/scientific_data/LICENCE_REMEDIATION.md`.

Quality tooling

Optional physics gates live in `spectro_scraper/quality.py` (^{13}C peak count – carbon count; ^1H integration vs formula; IR wavenumber windows). They were **not** applied as a hard filter to every released row at harvest; Technical Validation reports post-hoc checks below. Transcription fidelity uses `scripts/audit_extraction.py`.

Data Records

Object definition

Each IReXP record is a JSON object. Required chemistry fields for an IR entry:

Field	Type	Description
id	string	Stable record id (InChIKey when resolved, else internal)
ir_bands_cm-1	float list	Experimental IR peak positions (cm ⁻¹)
ir_source	string	experimental for all released rows
source_doi	string	PMC:<pmcid> or Chemotion deposit DOI
h_nmr / c_nmr	string or null	Author-reported shift lists
smiles / selfies / inchikey	string or null	Resolved structure encodings
has_structure	bool	Convenience flag
license / license_pool	string	Per-article stamp (Europe PMC join or Chemotion)

Not included: absorbance traces, intensities, instrument metadata beyond what appears in source text, PDFs, figures, or full article bodies.

Files and counts

Paths relative to the project repository / Hugging Face mirror.

File	Records	Description
data/irexp/irexp.jsonl.gz	121,233	Full curated release
data/irexp_resolved/irexp_resolved.jsonl.gz	43,060	100% structure-linked
data/irexp_release/train_no_bench.jsonl.gz	42,808	Resolved minus IRSpectra-Bench InChIKeys
data/irexp_release/train_no_bench_nmr.jsonl.gz	32,949	Same with both ¹ H and ¹³ C
data/irexp_release/pretrain_ir.jsonl.gz	119,345	PMC-only IR pretrain pool
data/irexp/seen_papers.txt.gz	188,016 lines	PMC IDs scanned at harvest
data/irexp/ir_harvest_snapshot.jsonl.gz	134,893	Intermediate harvest snapshot
data/irexp/licence_pools/irexp_commercial.jsonl.gz	87,617	CC-BY + CC0 (Zenodo/Sci Data primary)
data/irexp/licence_pools/irexp_non_commercial.jsonl.gz	129,038	NC* held aside
data/irexp/licence_pools/irexp_sharealike.jsonl.gz	1,897	Chemotion + rare PMC SA
data/irexp/licence_pools/irexp_empty_unknown.jsonl.gz	40,781	Excluded from commercial

Composition of irexp.jsonl.gz:

Slice	Count
All IR band-list records	121,233
With ^1H and/or ^{13}C NMR	87,075 (72%)
With resolved structure	43,060 (35.5%)
Structure + any NMR	40,702
Full IR + ^1H + ^{13}C + structure	33,201
PMC OA provenance	119,345
Chemotion provenance	1,888
Unique PMC accessions	15,416

Licence-pool counts (confirmed Track 1 join, 2026-08-26). Lookup over **15,416** unique PM-CIDs; stamped pool files under `data/irexp/licence_pools/` (see `docs/scientific_data/LICENCE_REMEDIATION.md` and `data/irexp/pmc_licence_summary.json`). Sum of pools = **121,233** (unchanged).

Pool file	Count	Notes
<code>irexp_commercial.jsonl.gz</code>	87,617	CC-BY / CC0 — Zenodo primary
<code>irexp_non_commercial.jsonl.gz</code>	20,938	CC-BY-NC* held aside
<code>irexp_empty_unknown.jsonl.gz</code>	10,781	empty / unresolved — excluded from commercial deposit
<code>irexp_other.jsonl.gz</code>	0	reserved (e.g. CC-BY-ND)
<code>irexp_sharealike.jsonl.gz</code>	1,897	Chemotion CC-BY-SA-4.0 (1,888) + rare PMC SA

The full `irexp.jsonl.gz` remains multi-licence on disk; commercial redistribution must use the commercial pool (or `license_pool == "commercial"`). Hugging Face should be re-issued with these files (`scripts/publish_hf.py`).

Median bands: **9** (PMC), **39** (Chemotion). All **1,360,866** released IR band values fall inside 350–4000 cm^{-1} (full-corpus range check; Technical Validation).

Access

- **Hugging Face:** <https://huggingface.co/datasets/ilkhamfy/IRexp> (public mirror; re-issue with stamped pools — see `LICENCE_REMEDIATION.md`).
- **GitHub development mirror:** <https://github.com/IlkhamFY/spectro-agent>
- **Zenodo archival snapshot:** [TODO: 10.5281/zenodo.XXXXXXX] (mint after honest multi-licence metadata).

Technical Validation

Transcription fidelity (PMC)

On a seed-fixed random sample of **60** PMC-sourced records (`scripts/audit_extraction.py --n 60 --seed 0`), each article was re-fetched from Europe PMC and every recorded wavenumber was checked against the source text ($\pm 1 \text{ cm}^{-1}$ integer match). Result: **560/560 bands** and **60/60**

records confirmed (Wilson 95% CI 99.3–100% for bands; 94–100% for records). Frozen machine-readable summary: `data/audit/extraction_audit.json`.

This bounds **transcription** error (hallucinated, mis-parsed, or unit-mangled values). It does **not** measure extraction **recall** (whether every IR string in every paper was found) — that requires human reading of full experimental sections and remains planned (see below).

Structure–NMR consistency (resolved sample)

On a random sample of **500** `irexp_resolved` records with ^{13}C text (seed 0), comma-separated ^{13}C peaks were counted against RDKit carbon counts: **17/500 (3.4%)** listed more peaks than carbons (physically impossible $\rightarrow$ name/structure assignment or parse error). On **500** records with ^1H text, summed reported integrals exceeded formula hydrogens by more than 2 in **17/497** scored rows (**3.4%**; 3 lacked parseable integrals). Reproducible summary: `docs/scientific_data/qc_structure_nmr.json`.

These rates are reported as integrity diagnostics for re-users (quarantine or filter before supervised training). They are **not** claimed as a full expert structure audit; NMRexp-scale manual skeleton checks (n 300) remain future work.

IR physical window

Every band in the full 121,233-record release lies in **[350, 4000] cm^{-1}** (0 / 1,360,866 out of range).

Planned / deferred QC

Check	Status
Per-PMCID licence join + pool counts	Done — <code>LICENCE_REMEDIATION.md</code> ; commercial 87,617
Extraction-recall human audit (papers, not bands)	Planned (n 30–50 papers)
Larger transcription sample (n 200)	Planned
Full-corpus <code>quality.py</code> quarantine pass	Planned
Expert structure spot-check (n 100)	Optional / deferred

Usage Notes

- **Band lists spectra.** Do not evaluate models trained on IRexp as if they had seen full absorbance curves.
- **Separate pools by density and licence.** PMC (sparse, author-transcribed) vs Chemotion (denser, deposit peak-picked). Keep Chemotion ShareAlike constraints in mind when combining pools; combined redistribution of Chemotion-derived rows must honour CC-BY-SA-4.0.
- **Do not assume PMC = CC-BY-4.0.** Filter to `license_pool == "commercial"` (or use `irexp_commercial.jsonl.gz`) for commercial redistribution; attribute via `source_doi / pmcid`.
- **Training without benchmark leakage.** If using the complementary IRSpectra-Bench problems, fine-tune from `train_no_bench.jsonl.gz` (or rebuild with `scripts/build_train_no_bench.py`) so benchmark InChIKeys are withheld.

- **Structure coverage.** Prefer `irexp_resolved` for supervised structure tasks; 64.5% of records lack SMILES.
- **Attribution.** Cite this Data Descriptor / Zenodo DOI and attribute originating articles through each record's `source_doi`.

Data Availability

IRexp numeric extracts are available at:

- Hugging Face Datasets: <https://huggingface.co/datasets/ilkhamfy/IRexp>
- GitHub: <https://github.com/IlkhamFY/spectro-agent> (`data/irexp/`, `data/irexp_resolved/`, `data/irexp_release/`)
- Zenodo: [TODO: 10.5281/zenodo.XXXXXXX] (archival DOI at proof; primary artifact = commercial pool, cc-by-4.0 metadata + SA companion)

Licensing summary (honest):

- **Chemotion (1,888):** CC-BY-SA-4.0¹⁰.
- **PMC (119,345):** mixed Creative Commons — stamped per article; commercial redistributable **87,617** (CC-BY/CC0); NC* **20,938** held aside; empty/unknown **10,781** excluded from commercial Zenodo (`LICENCE_REMEDIATION.md`).
- Only extracted numeric fields and identifiers are redistributed; source full texts are not.

Code Availability

Harvesting, extraction, structure resolution, licence pool splitting, and validation scripts are in <https://github.com/IlkhamFY/spectro-agent> under the **MIT License** (`LICENSE`). Principal modules: `spectro_scraper/` (`extract.py`, `normalize.py`, `pipeline.py`, `quality.py`); validation scripts/`audit_extraction.py`; pool split scripts/`split_license_pools.py`. The version corresponding to this Descriptor will be tagged at Zenodo deposit time.

Author contributions

I.Y.: conceptualization, methodology, software, data curation, validation, writing — original draft.

R.S.: methodology, software, validation, writing — review and editing.

R.A.V.-H.: conceptualization, supervision, writing — review and editing.

Competing interests

The authors declare no competing interests.

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References are maintained in `docs/scientific_data/references.bib` (pandoc/CSL build) and overlap the project bibliography where shared.

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